Mouse database v2: what data is available and what is in it

`data_exploration.ipynb` (v1) describes the database but analyses only trail 3. This notebook looks at **all trails** and at **both exports**: the 2023 pickle `/data/mice/maus.pandas.pkl` (converted from `maus_mysql.sql`) and the phpMyAdmin dump `mausprojekt-3.sql` of 2025-08-08, converted to `/data/mice/mausprojekt-3.pandas.pkl` with `tfmplayground/mouse/sql_dump.py`.

- **Part A** compares the two exports and decides which one is the base.
- **Part B** describes all trails of the new export.
- **Part C** checks how usable each trail is for weight forecasting.
- **Part D** looks at the new tables (lab samples, the `Logbuch`) and at columns v1 never used.
- **Part E** summarises data quality and lists candidate datasets for modelling.

No models are trained here. Every number in the text cells is computed in the notebook.

0. Setup

The worktree that holds this notebook comes first on `sys.path`, so `tfmplayground` (including the new `sql_dump.py`) is imported from here and not from the editable install of the main checkout.

```
tfmplayground imported from /workspaces/nanoPFN/src/TFM-Playground/.claude/worktree
s/bridge-cse_01Kxsthj9LqeaqKaNs4ZbzvC/tfmplayground/__init__.py
```

English names for the German lookup tables

Names are read from the tables (`Rollen`, `Versuchsreihen`, `Tierversuche`, `Todesursachen`, ...). v1's English translations are used where they exist; new entries fall back to the German text.

Plot style and helpers (from v1)

Tidy frames

v1's setup cells, wrapped in `build_frames(tables)` so that they run on both exports: `mice`, `visits`, `radio`, `chemo`, `surgery`, `anaesthesia`, `tumours`, plus the lookups.

Load both exports

The old pickle names the long-text columns `Operationen.OpBericht` and `Prozeduren.Befund` "`longtext`" and keeps their `\r\n` escaped (a quirk of the 2023 conversion, see `sql_dump.py`). For the comparison the old tables get the real names and unescaped text; the loaders read the pickles unchanged.

```
old: 25 tables, 843 mice, 22142 visits, 15 trails
new: 55 tables, 988 mice, 28125 visits, 18 trails, last visit 2025-06-24
```

Part A: the two exports side by side

A1. Table inventory

Rows and columns of every table in both exports.

	table	category	rows_old	rows_new	columns_old	columns_new	
0	Visiten	core	22142	28125	41	41	c weig s
1	Versuchstiere	core	843	988	29	29	c m s dat
2	Narkosen	treatment	2823	3114	10	10	a
3	Bestrahlungen	treatment	1958	2158	18	18	c ra
4	Chemotherapie	treatment	1316	1406	13	13	c che adm
5	Operationen	treatment	47	67	9	9	(t
6	Prozeduren	treatment	13	13	7	7	
7	Tumor	outcome	844	973	9	9	tu m in ren tum
8	VGruppe	design	66	75	6	6	
9	Kohorten	design	27	31	6	6	coh
10	Versuchsreihen	design	15	18	15	15	tr
11	Tierversuche	design	12	15	11	11	pr

	table	category	rows_old	rows_new	columns_old	columns_new	
12	Tumorzelllinien	lookup	19	21	6	6	lo
13	RTKonzepte	lookup	20	20	10	10	lo
14	ChTKonzepte	lookup	18	19	15	15	lo
15	Topographie	lookup	15	15	4	4	lo
16	Rollen	lookup	11	14	3	3	lo
17	Tiertypen	lookup	9	10	7	7	lo
18	Todesursachen	lookup	8	8	3	3	lo
19	ChemoMed	lookup	6	7	7	7	lo
20	Applikation	lookup	6	6	3	3	lo
21	Toetungsarten	lookup	5	5	3	3	lo
22	SeitenBez	lookup	4	4	2	2	lo
23	Zuechter	lookup	4	4	6	6	lo
24	Tagescheck	admin	213	232	14	14	da ch
25	Gewebe	lab/sample	<NA>	7362	<NA>	10	tissu (To
26	Paraffin	lab/sample	<NA>	4500	<NA>	12	sa a
27	Kryo	lab/sample	<NA>	3733	<NA>	15	cryo sa a
28	PCR	lab/sample	<NA>	3134	<NA>	20	r cou b ma
29	Knochenmark	lab/sample	<NA>	1034	<NA>	9	bo s m

	table	category	rows_old	rows_new	columns_old	columns_new	
30	Blut	lab/sample	<NA>	1028	<NA>	9	bloc da
31	Druckmessungen	lab/sample	<NA>	2	<NA>	6	mea (r
32	MRTSequenzen	lab/sample	<NA>	2	<NA>	8	MRI
33	Objekttraeger	lab/sample	<NA>	1	<NA>	9	r s
34	PCR_Import	lab/sample	<NA>	0	<NA>	18	st
35	Logbuch	logging	<NA>	176922	<NA>	6	log a lo
36	Organe	lookup	<NA>	33	<NA>	4	lo
37	KlinikFirma	lookup	<NA>	15	<NA>	10	lo
38	Organsysteme	lookup	<NA>	12	<NA>	3	lo
39	Markierungen	lookup	<NA>	11	<NA>	5	lo
40	Farben	lookup	<NA>	9	<NA>	3	lo
41	Faerbungen	lookup	<NA>	8	<NA>	4	lo
42	Qualifikationen	lookup	<NA>	7	<NA>	3	lo
43	MRTWichtungen	lookup	<NA>	5	<NA>	3	lo

	table	category	rows_old	rows_new	columns_old	columns_new	
44	Traeger	lookup	<NA>	4	<NA>	3	lo
45	Messkammern	lookup	<NA>	3	<NA>	7	lo
46	MRTKontrast	lookup	<NA>	2	<NA>	4	lo
47	OTBoxTypen	lookup	<NA>	2	<NA>	6	lo
48	Standorte	lookup	<NA>	2	<NA>	5	lo
49	OTBoxen	lookup	<NA>	1	<NA>	7	lo
50	OTSchnittKonzept	lookup	<NA>	1	<NA>	4	lo
51	OTSchnittProzeduren	lookup	<NA>	1	<NA>	6	lo
52	Module	admin	<NA>	61	<NA>	10	m
53	SystemVariablen	admin	<NA>	14	<NA>	3	co
54	Destinationen	admin	<NA>	2	<NA>	3	tar

```
25 shared tables, 30 only in the new export, 0 only in the old one
Shared tables with different columns: none
Visiten columns in the dump (before conversion): 40 | 'Zeitpunkt' in the dump: Fals
e | in the old dump: False
```

`Visiten.Zeitpunkt` is in neither dump: the 2023 conversion derived it from `Visitendatum` and `Visitenzeit`, and `sql_dump.py` derives it the same way (its self-check compares it with the old pickle). Neither loader reads it:

```
'Zeitpunkt' used by michael.py or preprocessing.py: False
```

A2. Is the old export a subset of the new one?

Every shared table is joined on its primary key (the first column, checked to be unique in both). A row is *unchanged* if all shared columns are equal (missing = missing), *changed* otherwise, *deleted* if its key is only in the old export and *added* if only in the new one.

	key	rows_old	rows_new	unchanged	changed	deleted	add
Applikation	IdApplikation	6	6	6	0	0	
Bestrahlungen	IdBestrahlung	1958	2158	1958	0	0	2
ChTKonzepte	IdChTKonzept	18	19	18	0	0	
ChemoMed	IdChemoMed	6	7	6	0	0	
Chemotherapie	IdChemo	1316	1406	1312	4	0	
Kohorten	IdKohorte	27	31	27	0	0	
Narkosen	IdNarkose	2823	3114	2823	0	0	2
Operationen	IdOp	47	67	47	0	0	
Prozeduren	IdProzedur	13	13	13	0	0	
RTKonzepte	IdRTKonzept	20	20	20	0	0	
Rollen	IdRolle	11	14	11	0	0	
SeitenBez	IdSeite	4	4	4	0	0	
Tagescheck	IdCheck	213	232	213	0	0	
Tiertypen	IdTiertyp	9	10	9	0	0	
Tierversuche	IdVersuch	12	15	12	0	0	
Todesursachen	IdTodesursache	8	8	8	0	0	
Toetungsarten	IdToetungsart	5	5	4	1	0	
Topographie	IdTopographie	15	15	15	0	0	
Tumor	IdTumor	844	973	844	0	0	1
Tumorzelllinien	IdZelllinie	19	21	19	0	0	
VGruppe	IdVGruppe	66	75	62	4	0	
Versuchsreihen	IdVersuchsreihe	15	18	13	2	0	
Versuchstiere	IdTier	843	988	803	40	0	1
Visiten	IdVisite	22142	28125	22142	0	0	59
Zuechter	IdZuechter	4	4	4	0	0	

The changed rows in detail. In `Versuchstiere`, the change per mouse and trail:

trail	0.0	1.0	15.0	Total
change				
death date, cause and killing method added	0	0	38	38
other: AktivV	1	1	0	2
Total	1	1	38	40

	table	key	column	old	new
0	Chemotherapie	1286	IdApplikationIst	1.0	2
1	Chemotherapie	1300	IdApplikationIst	1.0	2
2	Chemotherapie	1311	IdApplikationIst	1.0	2
3	Chemotherapie	1312	IdApplikationIst	1.0	2
4	Toetungsarten	1	Toetungsart	C02	C02/O2
5	VGruppe	59	Zellen	PC-3 RGB + Luziferase	PC-3 RGB + Luziferase1
6	VGruppe	60	Zellen	PC-3 RGB + Luziferase	PC-3 RGB + Luziferase2
7	VGruppe	61	Zellen	PC-3 RGB + Luziferase	PC-3 RGB + Luziferase3
8	VGruppe	62	Zellen	PC-3 RGB + Luziferase	PC-3 RGB + Luziferase4
9	Versuchsreihen	11	NameReihe	Vorversuch PC3	Vorversuch N127/2019 PC3
10	Versuchsreihen	12	NameReihe	Hauptversuch PC3	Hauptversuch N127/2019 PC3

```
Operationen: 0 rows with an edited free-text column ()
Prozeduren: 0 rows with an edited free-text column ()
```

Does the loader see the same trail-3 data in both exports? `michael.load_dataset` with v1's configuration (regression check against v1: 186 mice, 3,049 rows).

```
trail 3, old export: 186 mice, 3049 rows
trail 3, new export: 186 mice, 3049 rows
The trail-3 study cohort is identical in both exports.
all trails, roles 1-3, old export: 434 mice, 12397 rows
```

A3. Which export is the base?

What this tells us (Part A).

- The new export has 55 tables instead of 25: all 25 old tables with the same columns, plus 30 new ones. `Visiten.Zeitpunkt` is a column the 2023 conversion added, not one the database lost; the loaders do not use it.
- **The old export is contained in the new one:** 0 rows deleted, 0 of 22,142 visits changed, 5,983 visits and 145 mice added. The edits to old rows are completions and relabelling: 38 mice of trail(s) 15 got their death date, cause and killing method (this changes what the loader keeps for trail 15), 2 mice changed their `AktivV` flag, and 11 values in other tables changed (Chemotherapie.IdApplikationIst 4, Toetungsarten.Toetungsart 1, VGruppe.Zellen 4, Versuchsreihen.NameReihe 2), i.e. chemotherapy routes from "unknown" to i.p. and renamed trails, batches and killing method.
- The trail-3 study cohort through `michael.load_dataset` is identical in both exports (186 mice, 3,049 rows), so v1 and the results stay valid on the new export.
- **Base for the rest of the notebook: the new export.**

Part B: all trails (new export)

B1. One row per trail

trail	name	licence	project	start	end	first_visit	last_visit
0	(not in Versuchsreihen)	NaN	NaN	NaN	NaN	2019-10-21	2019-11-25
1	Pilot 1	G14/13	Influence of RT, ChT, simultaneous RChT and surgery on metastasis in xenogra...	2013-06-17	None	2013-06-17	2016-10-24
2	Cell migration under irradiation	G133/12	Selectin-dependent interaction of cell populations in a clinically relevant ...	2013-07-24	None	2013-08-28	2013-10-16
3	Main experiment 1	G14/13	Influence of RT, ChT, simultaneous RChT and surgery on metastasis in xenogra...	2013-08-03	None	2013-06-17	2016-08-08
4	Pilot Loges 1	G36/13	Resistance and bone-marrow micrometastasis under anti-androgen therapy of br...	2013-11-27	None	NaN	NaN
5	Pilot 1 (methods)	G	Pilot with two animals: learning methods	None	None	NaN	NaN
6	Breast cancer model	NaN	NaN	2014-06-10	None	2014-10-27	2018-02-15
7	Nanoparticles	G133/12	Selectin-dependent interaction of cell populations in a clinically relevant ...	2015-08-27	None	2015-08-27	2015-11-16

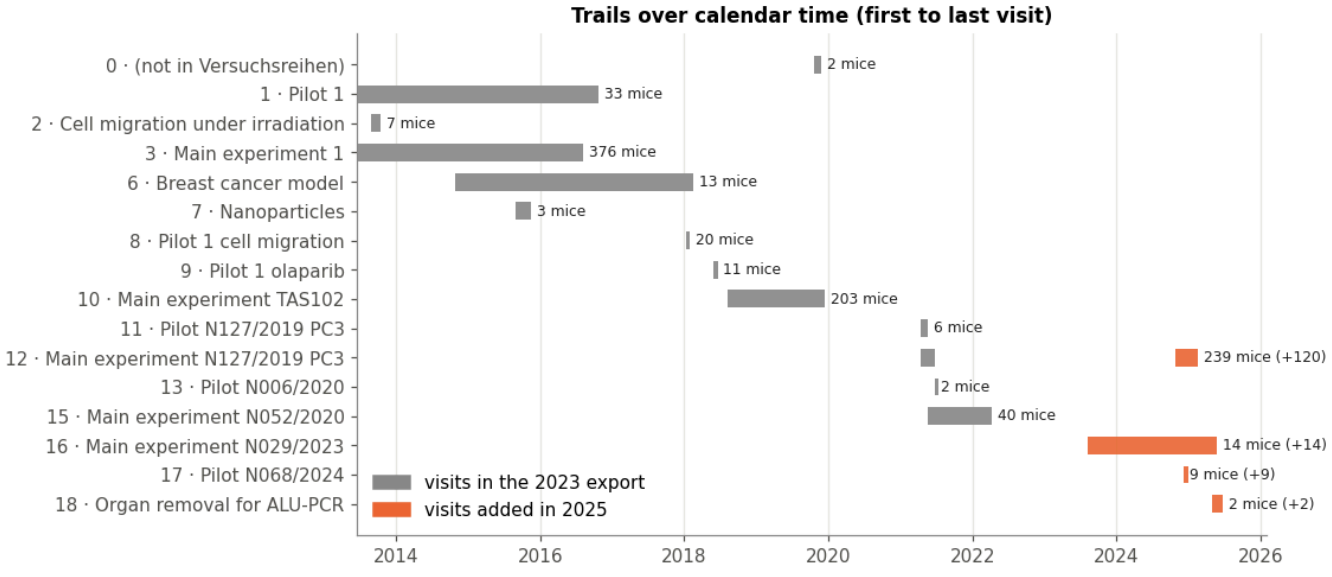
trail	name	licence	project	start	end	first_visit	last_visit
8	Pilot 1 cell migration	G14/13	Influence of RT, ChT, simultaneous RChT and surgery on metastasis in xenogra...	2018-01-08	None	2018-01-08	2018-01-29
9	Pilot 1 olaparib	G67/16	Radiosensitisation of PTEN- or CHD1-depleted prostate carcinomas by PARP1 in...	2018-05-31	None	2018-05-31	2018-06-14
10	Main experiment TAS102	N013/2018	TAS102	2018-05-01	None	2018-08-06	2019-12-16
11	Pilot N127/2019 PC3	N127/2019	Influence of locally ablative RT, ChT, RChT and surgery on metastasis in two...	2021-04-13	2022-04-13	2021-04-15	2021-05-21
12	Main experiment N127/2019 PC3	N127/2019	Influence of locally ablative RT, ChT, RChT and surgery on metastasis in two...	2021-04-16	None	2021-04-15	2025-02-16
13	Pilot N006/2020	N006/2020	Immunomodulation by radiotherapy in glioblastoma	2021-06-22	None	2021-06-22	2021-06-22
14	Main experiment N006/2020	N006/2020	Immunomodulation by radiotherapy in glioblastoma	2021-06-23	2021-06-23	NaN	NaN
15	Main experiment N052/2020	N052/2020	Relevance of clonal heterogeneity for cerebral tropism and radiotherapy resp...	2021-09-17	2021-10-17	2021-05-20	2022-04-07

trail	name	licence	project	start	end	first_visit	last_visit
16	Main experiment N029/2023	N029/2023	Tracking of systemic drugs, nanoparticles and immune cells in a xenograft pr...	2023-08-08	None	2023-08-08	2025-05-25
17	Pilot N068/2024	N068/2024	Functional role of small-RNA clusters in hepatocarcinogenesis	2024-11-22	None	2024-12-10	2024-12-10
18	Organ removal for ALU-PCR	ORG1148/2025	Organ removal as a supplement to project N029/2023	2025-04-23	None	2025-04-29	2025-06-24

B2. What is new since the 2023 export

Mice and visits per trail in both exports, and the trails over calendar time: grey is the time span of the visits already in the old export, orange that of the added visits.

trail	mice_old	mice_new	visits_old	visits_new	mice_added	visits_added	new trail
12	119	239	4705	10062	120	5357	False
16	0	14	0	501	14	501	True
17	0	9	0	9	9	9	True
18	0	2	0	116	2	116	True



B3. Protocols per trail

Radiotherapy concepts (`RTKonzepte` , from the fractions in `Bestrahlungen`) and chemotherapy drugs and concepts (from `Chemotherapie`), with the number of mice. The schedule is fractions x dose per fraction of the concept (the concepts also differ in beam energy, 200 vs 225 kV). The last columns show how much of each trail uses the same RT schedule and ChT drugs as trail 3.

trail	RT concepts (mice)	RT schedules (mice)	RT fractions per mouse	RT total dose per mouse (Gy)	ChT drugs (mice)	ChT concepts (mice)	ChT administrations per mouse
0	d1-5 2.0 Gy 1	5 x 2 Gy 1	3 [3–3]	9 [9–9]			
1	d1-5 10,0 Gy 8	5 x 10 Gy 8	5 [5–5]	50 [48–50]	Cisplatin 19, Etoposid 6, Docetaxel 3	BC Standard alt 9, Doctaxel 20 mg/kg d1-8-15 3, Cisplatin 6mg/kg d1 2, Cispl...	3 [3–4]
2	d1 10,0 Gy 6, d1 15,0 Gy 1	1 x 10 Gy 6, 1 x 15 Gy 1	1 [1–1]	10 [10–10]			
3	d1-5 10,0 Gy 134	5 x 10 Gy 134	5 [5–5]	50 [50–50]	Cisplatin 76, Docetaxel 40	Cisplatin 6mg/kg d1 76, Doctaxel 20 mg/kg d1-8-15 40	1 [1–3]
4	d1 6 Gy - Femora 3, d1 6 Gy - Mamma 3	2 x 6 Gy 3, 1 x 6 Gy 3	2 [1–2]	9 [6–12]			
6	Mamma02 d1 10,0 Gy 11, Mamma01 d1 3,0 Gy 1	1 x 10 Gy 11, 1 x 3 Gy 1	1 [1–1]	10 [10–10]			
7	d1-5 10,0 Gy 3	5 x 10 Gy 3	5 [5–5]	50 [50–50]			
8	d1-4 10.0 Gy 1, d1 10,0 Gy 1, d1-3 10.0 Gy 1, d1-5 10,0 Gy 1, d1-2 10.0 Gy 1	4 x 10 Gy 1, 1 x 10 Gy 1, 3 x 10 Gy 1, 5 x 10 Gy 1, 2 x 10 Gy 1	3 [2–4]	30 [20–40]			
9	d1 2.0 Gy 8	1 x 2 Gy 8	1 [1–1]	2 [2–2]	Olaparib 6	Olaparib 2h vor RT 6	1 [1–1]
10	d1-10 3,0 Gy 68, d1-5 2.0 Gy 40, d1 2.0	10 x 3 Gy 68, 5 x 2 Gy 40, 1 x	10 [3–10]	20 [9–30]	TAS102 70, Capecitabine 53	TAS102 d1-10 52, Capecitabine d1-10 27,	10 [5–10]

trail	RT concepts (mice)	RT schedules (mice)	RT fractions per mouse	RT total dose per mouse (Gy)	ChT drugs (mice)	ChT concepts (mice)	ChT administrations per mouse
	Gy 4, d1 3.0 Gy 4	2 Gy 4, 1 x 3 Gy 4				Capecitabine d1-5 26, TAS102 d1-5 18	
11	d1-5 10,0 Gy 225 kV 3, d1-5 10,0 Gy 2	5 x 10 Gy 5	1 [1–5]	10 [10–50]	Docetaxel 2	Docetaxel d1,8 4 mg/kg 1, Docetaxel D1 8mg/kg 1	2 [1–2]
12	d1-5 10,0 Gy 225 kV 69, d1-5 10,0 Gy 10	5 x 10 Gy 79	5 [5–5]	50 [50–50]	Docetaxel 76	Docetaxel d1,8 4 mg/kg 76	2 [2–2]
13	d1 10 Gy225 kV 2	1 x 10 Gy 2	1 [1–1]	10 [10–10]			
15	d1-10 2.0 Gy 10, d1-6 6x2,0 Gy 8	10 x 2 Gy 10, 6 x 2 Gy 8	10 [6–12]	19 [12–24]	Docetaxel 1	Docetaxel d1,8 4 mg/kg 1	2 [2–2]
16					Ruthenium-Glutamin 10	Ruthenium-Glutamin d1 10	1 [1–1]

IdRTKonzept	RTKonzept	dose per fraction (Gy)	total dose (Gy)	fractions	fractions/day	fractions/week	beam
1	d1-5 10,0 Gy	10	50	5	1	5	200 kV
2	d1 10,0 Gy	10	10	1	1	1	200 kV
3	d1 15,0 Gy	15	15	1	1	1	200 kV
4	d1 6 Gy - Mamma	6	6	1	1	1	200 kV
5	d1 6 Gy - Femora	6	12	2	2	2	200 kV
6	Mamma01 d1 3,0 Gy	3	3	1	1	1	200 kV
7	Mamma02 d1 10,0 Gy	10	10	1	1	1	200 kV
8	d1-4 10.0 Gy	10	40	4	1	4	200 kV
9	d1-3 10.0 Gy	10	30	3	1	3	200 kV
10	d1-2 10.0 Gy	10	20	2	1	2	200 kV
11	d1 2.0 Gy	2	2	1	1	1	200 kV
12	d1-10 3,0 Gy	3	30	10	1	5	200 kV
13	d1-5 2.0 Gy	2	10	5	1	5	200 kV
14	d1 3.0 Gy	3	3	1	1	1	200 kV
15	d1-10 2.0 Gy	2	20	10	1	5	200 kV
17	d1-5 10,0 Gy 225 kV	10	50	5	1	5	225 kV
18	d1 10 Gy225 kV	10	10	1	1	1	225 kV
19	d1-11 12x2,0 Gy	2	24	12	2	6	225 kV
20	d1-6 6x2,0 Gy	2	12	6	1	6	225 kV

IdChTKonzept	protocol	IdChTMed1	dose 1 (mg/kg)	days 1	IdChTMed2	dose 2 (mg/kg)	days 2	Bem
1	BC Standard alt	1	6	1	2.0	12.0	3	
2	Doctaxel 20 mg/kg d1-8-15	3	20	3	NaN	NaN	None	
3	Cisplatin Dosis 1/3	1	2	1	NaN	NaN	None	
4	Cisplatin Dosis 2/3	1	4	1	NaN	NaN	None	
5	Cisplatin 6mg/kg d1	1	6	1	0.0	0.0		
6	Etoposid Dosis 1/3	2	4	3	NaN	NaN	None	
7	Etoposid Dosis 2/3	2	8	3	NaN	NaN	None	
8	Etoposid Dosis 3/3	2	12	3	NaN	NaN	None	
9	Cisplatin + Pause + Etoposid	1	6	1	2.0	12.0	3	Cisp Etop
10	Cisplatin red. + Pause + Etoposid	1	4	1	2.0	12.0	3	Cisp Etop
12	Olaparib 2h vor RT	4	1	1	NaN	NaN	None	
13	TAS102 d1-10	5	150	10	NaN	NaN	None	1
14	Capecitabine d1-5	6	400	5	NaN	NaN	None	Cape
15	TAS102 d1-5	5	150	5	NaN	NaN	None	1
16	Capecitabine d1-10	6	400	10	NaN	NaN	None	Cape 40
17	Docetaxel d1,8 4 mg/kg	3	4	2	0.0	0.0	None	D d1,8
18	Docetaxel D1 8mg/kg	3	8	1	NaN	NaN	None	
19	Ruthenium-Glutamin d1	7	200	1	7.0	NaN	None	Rut Glut

B4. Roles 12–14 (new in 2025)

	meaning
12	OE: organ removal
13	MP: injection of macrophages
14	NP: injection of nanoparticles

role	MP	NP	OE	Total
trail				
16	3	1	0	4
18	0	0	2	2
Total	3	1	2	6

Rows per table for the mice of these roles:

	mice	Visiten	Bestrahlungen	Chemotherapie	Operationen	Narkosen	Tumor
role_abbr							
MP	3	97	0	0	0	0	3
NP	1	32	0	0	0	0	1
OE	2	116	0	0	0	0	0

	trail	role	visits	first	last	weight
id						
985	16.0	13.0	32	2025-04-29	2025-05-24	18.10
986	16.0	13.0	31	2025-04-29	2025-05-24	18.90
987	16.0	14.0	32	2025-04-29	2025-05-25	17.90
988	16.0	13.0	34	2025-04-29	2025-05-25	20.00
989	18.0	12.0	58	2025-04-29	2025-06-24	20.05
990	18.0	12.0	58	2025-04-29	2025-06-24	23.15

What this tells us (Part B).

- The new export has 18 trails (3 new: 16, 17, 18). Most of the growth is trail 12, which went from 119 to 239 mice and from 4,705 to 10,062 visits; the new trails are small (trail 16: 14 mice, trail 17: 9 mice, trail 18: 2 mice).
- Trail 3 remains the largest trail (376 mice); the only other trails with more than 100 mice are 10 (203), 12 (239).
- Trails in which at least half of the irradiated mice got trail 3's RT schedule (5 x 10 Gy): 1, 3, 7, 11, 12. Trails in which at least half of the ChT mice got a trail-3 drug (Cisplatin, Docetaxel): 1, 3, 11, 12, 15. The other trails ran different protocols (e.g. fractionated 2–3 Gy schedules, TAS102/capecitabine) and are different experiments.
- Roles 12–14 cover 6 mice in trail(s) 16, 18 with 245 visits (0.9% of all visits); their weights were recorded but they got no RT/ChT, and the organ-removal mice have necropsy samples only.

Part C: usability per trail for weight forecasting

C1. Cohort flow per trail

Replays `michael.load_dataset` step by step (as v1 C1), for all trails at once. Each column is the number of mice left after that filter. "lost to the trail-start filter" counts mice that would have had enough weights if the records before the trail start or birth were not dropped. The final column is checked against `michael.load_dataset`.

Roles RT/RChT/ChT — kept: 519 mice, 16,376 rows (old export: 434 mice, 12,397 rows)

trail	all mice	in Versuchsreihen	role in [1, 2, 3]	death date known	treated (RT/ChT/OP recorded)	≥ 5 weights after trail start	all weights > 0 g	all weights ≤ 100 g	lost to the trail-start filter
0	2	0	0	0	0	0	0	0	0
1	33	33	31	29	29	29	26	26	0
2	7	7	7	0	0	0	0	0	0
3	376	376	188	188	188	188	187	186	0
4	6	6	6	0	0	0	0	0	0
5	2	2	0	0	0	0	0	0	0
6	13	13	13	13	13	0	0	0	0
7	3	3	3	3	3	3	3	3	0
8	20	20	5	5	5	5	5	5	0
9	11	11	10	10	10	0	0	0	0
10	203	203	164	158	156	156	155	153	0
11	6	6	6	6	6	6	6	6	0
12	239	239	115	115	115	115	115	114	0
13	2	2	2	0	0	0	0	0	0
15	40	40	19	19	18	18	18	17	1
16	14	14	10	10	10	9	9	9	0
17	9	9	0	0	0	0	0	0	0
18	2	2	0	0	0	0	0	0	0

Mice kept per trail and role:

role trail	ChT	RChT	RT	Total
1	18	7	1	26
3	53	60	73	186
7	0	0	3	3
8	0	0	5	5
10	44	71	38	153
11	1	1	4	6
12	36	39	39	114
15	0	0	17	17
16	9	0	0	9
Total	161	178	180	519

C2. Widening the role filter

The same flow with all roles. Controls and other untreated mice are then removed by the *treated* filter (Michael's loader drops mice without RT/ChT/OP), so a second run switches that filter off (`michael.UNTREATED` patched to an empty list, inside this notebook only) to see what the untreated mice could add as context.

All roles: 740 mice kept with the treatment filter, 888 mice and 26,974 rows without it.

trail	all mice	in Versuchsreihen	role in all	death date known	treated (RT/ChT/OP recorded)	≥ 5 weights after trail start	all weights > 0 g	all weights ≤ 100 g	lost to the trail-start filter
0	2	0	0	0	0	0	0	0	0
1	33	33	33	31	31	31	28	28	0
2	7	7	7	0	0	0	0	0	0
3	376	376	376	376	310	290	289	288	0
4	6	6	6	0	0	0	0	0	0
5	2	2	2	2	0	0	0	0	0
6	13	13	13	13	13	0	0	0	0
7	3	3	3	3	3	3	3	3	0
8	20	20	20	20	20	20	20	20	0
9	11	11	11	11	11	0	0	0	0
10	203	203	203	195	162	162	161	159	0
11	6	6	6	6	6	6	6	6	0
12	239	239	239	239	199	199	199	198	0
13	2	2	2	0	0	0	0	0	0
15	40	40	40	40	28	28	28	27	1
16	14	14	14	14	10	9	9	9	0
17	9	9	9	0	0	0	0	0	0
18	2	2	2	2	2	2	2	2	0

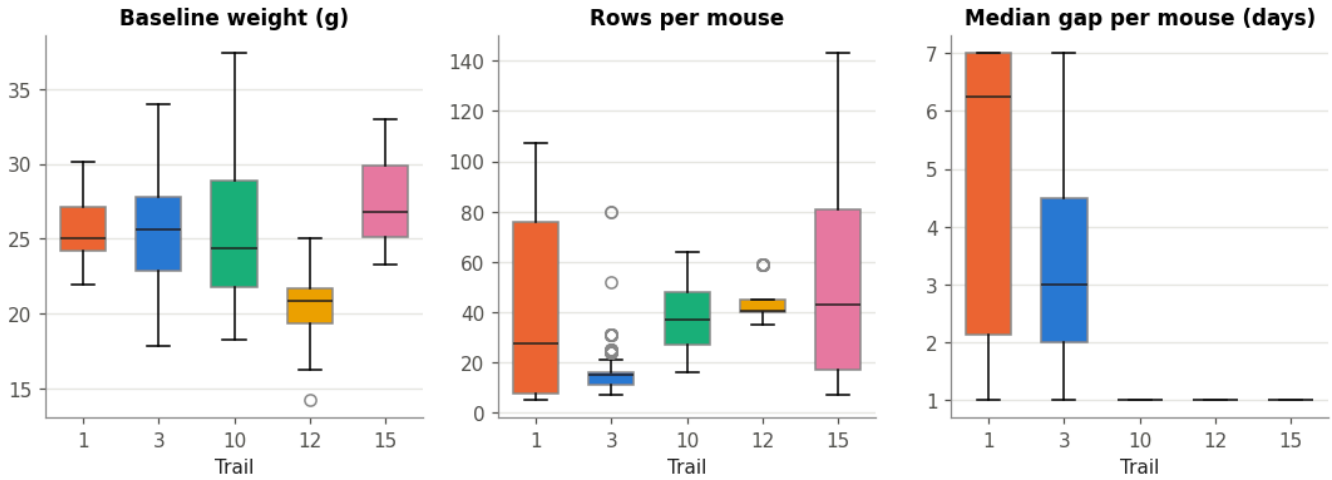
	mice	rows	with_treatment	series_length	median_gap_days	baseline_weight_g
0 –	5	150	0	30 [30–30]	1 [1–1]	31.2 [30.0–32.3]
1 RT	180	5761	180	31 [16–41]	1 [1–2]	23.5 [21.4–26.3]
2 RChT	178	5812	178	34 [16–43]	1 [1–2]	23.6 [21.1–27.2]
3 ChT	163	4838	161	27 [16–40]	1 [1–3]	24.2 [21.1–27.1]
4 K	86	1914	0	18 [8–32]	1 [1–6]	22.8 [20.9–25.4]
5 E	160	3908	2	22 [14–36]	1 [1–3]	23.6 [21.5–27.1]
7 B	6	208	0	22 [22–45]	2 [1–6]	25.8 [24.4–26.6]
8 KT	42	1869	0	46 [13–74]	1 [1–7]	25.5 [22.7–28.7]
9 OP	62	2294	62	37 [31–42]	1 [1–1]	21.2 [20.3–23.2]
12 OE	2	114	0	57 [57–57]	1 [1–1]	11.5 [11.3–11.7]
13 MP	3	79	0	26 [26–26]	1 [1–1]	12.5 [11.9–13.1]

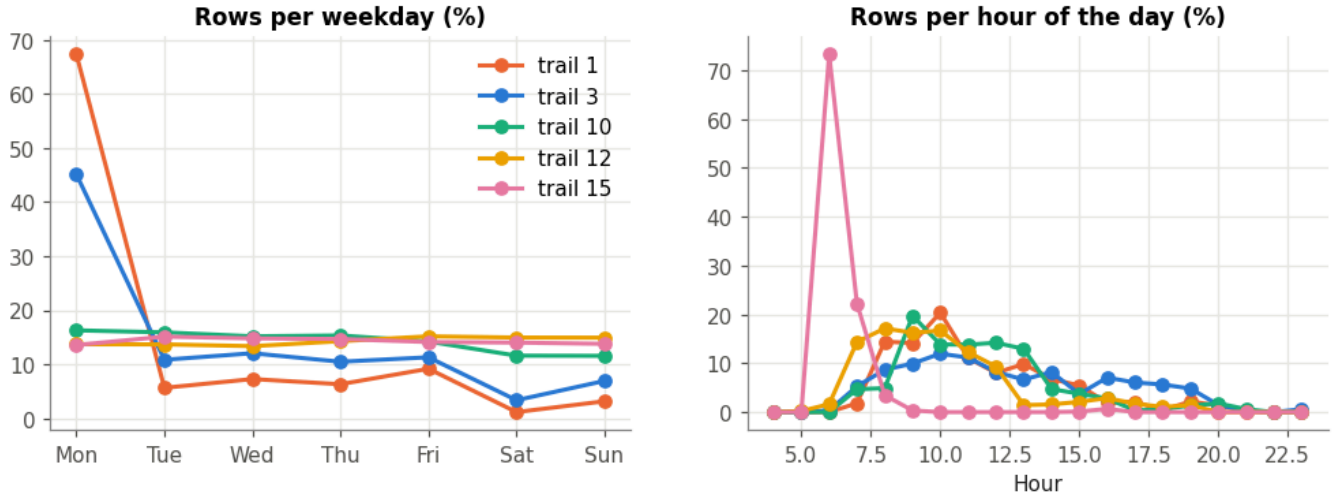
	mice	rows	with_treatment	series_length	median_gap_days	baseline_weight_g
14 NP	1	27	0	27 [27–27]	1 [1–1]	15.0 [15.0–15.0]

C3. Series characteristics per trail

For the roles RT/RChT/ChT through `michael.load_dataset` (`D13`), all trails. Gaps are days between consecutive rows of a mouse; loss is the lowest weight relative to the first one.

trail	mice	rows	series length	median gap (days)	% gaps of 1 day	% gaps of 7 days	% gaps > 7 days	% rows on Mondays	time of day (h)	% rows without time	baseline weight (g)
1	26	1047	28 [8–76]	7.0	15	56	4	67	10.9 [9.6–13.3]	0	25.1 [24.2–27.1]
3	186	3049	15 [11–16]	2.0	44	24	1	45	12.2 [10.1–16.1]	0	25.6 [22.8–27.8]
7	3	39	13 [13–13]	4.5	31	0	23	15	9.6 [7.5–15.5]	0	21.3 [21.2–21.8]
8	5	110	22 [22–22]	1.0	95	0	0	18	9.5 [8.4–10.4]	0	24.1 [23.7–25.4]
10	153	5808	37 [27–48]	1.0	93	0	0	16	11.7 [9.8–13.2]	0	24.4 [21.8–28.9]
11	6	181	29 [29–33]	1.0	95	0	0	14	8.8 [8.2–10.4]	0	21.1 [20.2–23.1]
12	114	4900	40 [40–45]	1.0	98	0	0	14	10.0 [8.5–11.6]	0	20.8 [19.3–21.7]
15	17	900	43 [17–81]	1.0	98	0	0	14	6.9 [6.8–7.0]	0	26.8 [25.1–29.9]
16	9	342	47 [5–62]	1.0	97	0	0	14	9.6 [8.1–11.2]	0	23.0 [20.9–24.3]

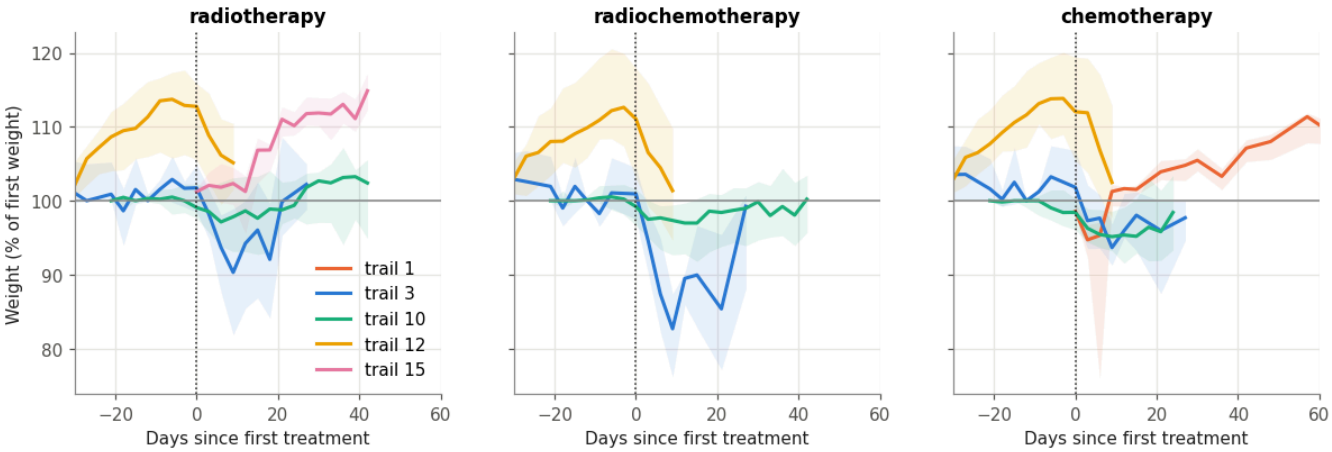




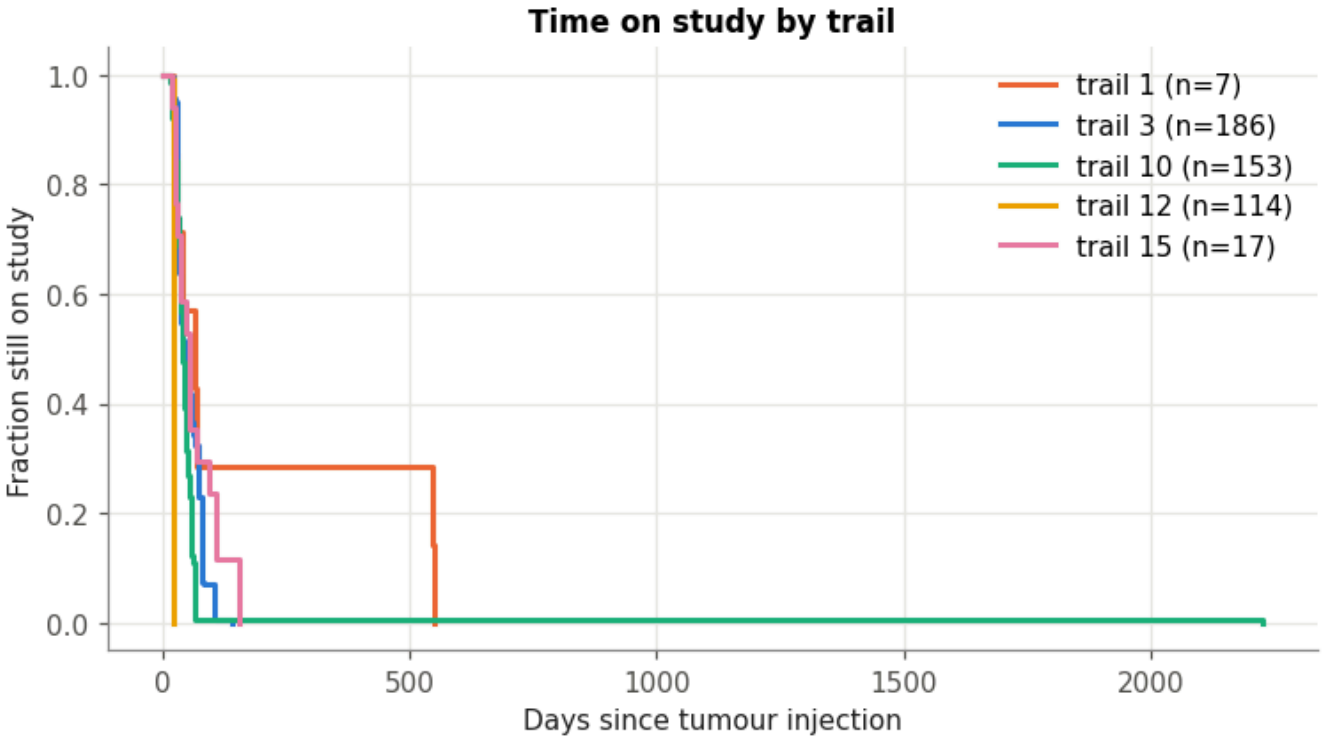
C4. Shift against trail 3

Population, era and protocol of the trails with at least `MIN_MICE` kept mice, and the weight course aligned at the first treatment (median over mice in 3-day bins, as v1 C4).

trail	mice	strain	% male	cell lines	years	roles	RT concepts	ChT drugs	days first row → first treatment
1	26	SCID 23, PFP/RAG2 3	81	PC-3 RGB 3, OH-1 RGB 3, H69 RGB 2	2013–2013	ChT 18, RChT 7, RT 1	d1-5 10,0 Gy 8	Cisplatin 16, Etoposid 6, Docetaxel 3	0 [0–21]
3	186	SCID 128, PFP/RAG2 58	66	OH-1 RGB 81, LNCaP RGB 58, H69 RGB 47	2013–2016	RT 73, RChT 60, ChT 53	d1-5 10,0 Gy 133	Cisplatin 74, Docetaxel 40	28 [21–49]
10	153	PFP/RAG2 153	43	SW48 55, HT29 49, HCT116 49	2018–2019	RChT 71, ChT 44, RT 38	d1-10 3,0 Gy 64, d1-5 2.0 Gy 37, d1 2.0 Gy 4, d1 3.0 Gy 4	TAS102 67, Capecitabine 48	14 [14–16]
12	114	SCID 114	100	PC-3DR RGB + Luziferase 60, PC-3 RGB + Luziferase 54	2021–2024	RChT 39, RT 39, ChT 36	d1-5 10,0 Gy 225 kV 69, d1-5 10,0 Gy 9	Docetaxel 75	32 [31–36]
15	17	NMRI-Foxn1nu 17	0	NCH644OBC 17	2021–2022	RT 17	d1-10 2.0 Gy 10, d1-6 6x2,0 Gy 7	NaN	0 [0–1]



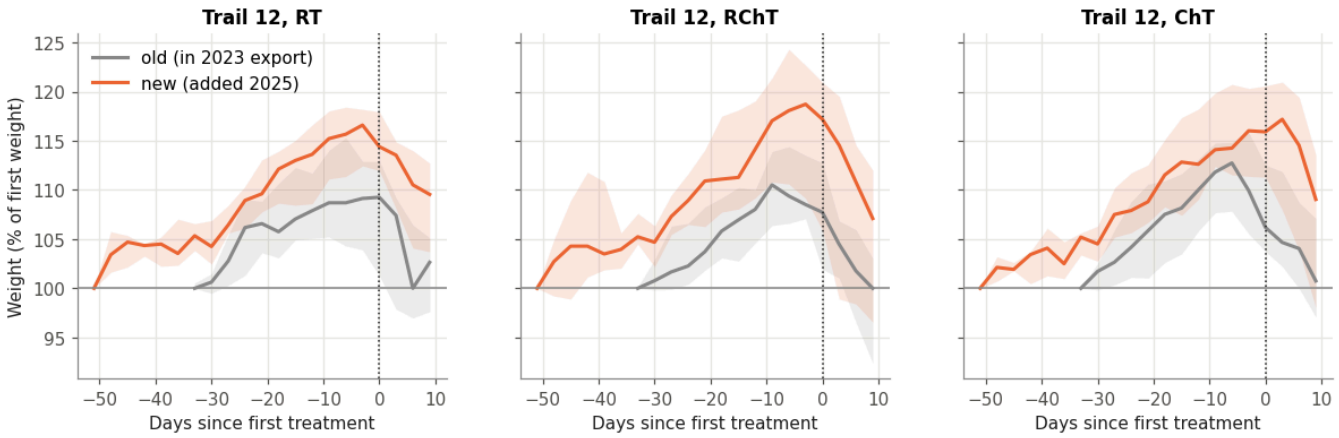
Time on study (tumour injection to death) per trail, RT/RChT/ChT mice of the kept series (v1's Kaplan–Meier plot, by trail instead of role). As in v1, almost all deaths are planned, so the curves show the follow-up design.



C5. Trail 12: old half vs new half

Trail-12 mice that are already in the 2023 export against those added since.

half	new (added 2025)	old (in 2023 export)
mice	120	119
first_visit	2024-10-24 00:00:00	2021-04-15 00:00:00
last_visit	2025-02-16 00:00:00	2021-06-25 00:00:00
cohorts	PC-3DR RGB + Luziferase Hauptversuch 120	PC-3 RGB + Luziferase Hauptversuch 119
cell_lines	PC-3DR RGB + Luziferase 120	PC-3 RGB + Luziferase 119
strains	SCID 120	SCID 119
roles	K 20, ChT 20, RChT 20, OP 20, RT 20, E 19, KT 1	OP 24, K 20, E 20, RChT 20, RT 19, ChT 16
male	120	119
RT concepts	d1-5 10,0 Gy 225 kV 40	d1-5 10,0 Gy 225 kV 29, d1-5 10,0 Gy 10
ChT drugs	Docetaxel 40	Docetaxel 36
kept series (RT/RChT/ChT)	60	54
series length	42 [40–48]	40 [40–41]
baseline weight (g)	20.9 [19.3–22.0]	20.8 [19.6–21.6]
lowest weight (% of first)	100.0 [95.8–100.0]	96.9 [93.7–100.0]
median gap (days)	1.0	1.0



C6. Naive last-value error per trail

The error of "next weight = last weight" on the rolling one-step-ahead origins (v1 C7), a yardstick for how hard each trail is. *Overall* is the mean of the three role MAEs, as in the results tables. The rolling machinery is run once on trail 3 and must match v1 (0.695 g); per

trail the same numbers are computed directly (the last-value forecast of origin k is weight $k - 1$).

```
Trail 3 via rolling_predictions: overall MAE 0.695 g
```

trail	mice	origins	MAE RT	MAE RChT	MAE ChT	MAE overall (mean of roles)	MAE all origins
1	26	995	0.996	0.536	0.561	0.697	0.564
3	186	2677	0.541	0.844	0.701	0.695	0.662
7	3	33	0.412	NaN	NaN	NaN	0.412
8	5	100	0.363	NaN	NaN	NaN	0.363
10	153	5502	0.147	0.172	0.175	0.165	0.166
11	6	169	0.450	0.484	0.500	0.478	0.466
12	114	4672	0.343	0.372	0.376	0.363	0.363
15	17	866	0.509	NaN	NaN	NaN	0.509
16	9	324	NaN	NaN	0.276	NaN	0.276

What this tells us (Part C).

- With v1's filters the new export yields 519 mice / 16,376 rows for RT/RChT/ChT over all trails (old export: 434 / 12,397); trails with at least 10 mice: 1 (26), 3 (186), 10 (153), 12 (114), 15 (17). Largest losses: role in [1, 2, 3] (407 mice), ≥ 5 weights after trail start (24 mice), death date known (23 mice).
- Widening the roles with Michael's treatment filter kept gives 740 mice (+221), because controls have no treatment. Without that filter 888 mice / 26,974 rows have series with ≥ 5 weights (369 more, 246 of them controls K/E), usable as untreated context.
- The trails are **different populations**: of the other trails with ≥ 10 mice, 3 share a mouse strain and 1 share a cell line with trail 3; cadence, series length and treatment timing differ too (tables above). Pooling is possible technically but mixes experiments.
- Trail 12's added mice use cell line(s) PC-3DR RGB + Luziferase against PC-3 RGB + Luziferase in the old half (a different tumour model, the same RT schedule and ChT drug, visits 2021-04–2021-06 vs 2024-10–2025-02). Baseline weight 20.8 [19.6–21.6] vs 20.9 [19.3–22.0] g, lowest weight 96.9 [93.7–100.0] vs 100.0 [95.8–100.0] % of the first weight (old vs new half). It is a second experiment under one trail id and should carry its own label.
- Naive last-value MAE: trail 3 0.662 g over all origins (0.695 g as the mean of the role MAEs, as in v1); other trails range from 0.166 to 0.564 g.

Part D: new tables and unused columns

D1. The 30 tables that are new in 2025

For every new table: purpose, rows, whether it refers to mice (`IdTier`), how many mice and trails it covers and the range of its first date column.

table	group	purpose	rows	columns	key to mice	mice	% of all mice	tra
Gewebe	lab/sample	tissue taken at necropsy: organ (Topographie), side, date, weight	7362	10	IdTier	813	82.3	0 3 7 9,
Paraffin	lab/sample	paraffin-embedded samples: sample type and storage position	4500	12	IdTier	813	82.3	0 3 7 9,
Kryo	lab/sample	cryo-preserved samples: sample type and storage position	3733	15	IdTier	806	81.6	0 3 7 9,
PCR	lab/sample	ALU-PCR results: cell counts and Cp values per blood/bone-marrow/tissue sample	3134	20	IdTier	641	64.9	0 3
Knochenmark	lab/sample	bone-marrow samples per mouse: date, running number, remark	1034	9	IdTier	802	81.2	0 3 8
Blut	lab/sample	blood samples per mouse: date, running number, remark	1028	9	IdTier	802	81.2	0 3 8

table	group	purpose	rows	columns	key to mice	mice	% of all mice	tra
Druckmessungen	lab/sample	pressure measurements (mmHg) per mouse and site	2	6	IdTier	1	0.1	
MRTSequenzen	lab/sample	MRI sequences per mouse	2	8	IdTier	2	0.2	1
Objekttraeger	lab/sample	microscope slides from paraffin blocks: staining, metastasis counts	1	9	IdTier	1	0.1	
PCR_Import	lab/sample	staging table for PCR imports (empty)	0	18	IdTier	0	0.0	
Logbuch	logging	log of the web application: logins, page views and submitted forms	176922	6		<NA>	NaN	
Organe	lookup	lookup table	33	4		<NA>	NaN	
KlinikFirma	lookup	lookup table	15	10		<NA>	NaN	
Organsysteme	lookup	lookup table	12	3		<NA>	NaN	
Markierungen	lookup	lookup table	11	5		<NA>	NaN	
Farben	lookup	lookup table	9	3		<NA>	NaN	
Faerbungen	lookup	lookup table	8	4		<NA>	NaN	
Qualifikationen	lookup	lookup table	7	3		<NA>	NaN	
MRTWichtungen	lookup	lookup table	5	3		<NA>	NaN	
Traeger	lookup	lookup table	4	3		<NA>	NaN	
Messkammern	lookup	lookup table	3	7		<NA>	NaN	
MRTKontrast	lookup	lookup table	2	4		<NA>	NaN	
OTBoxTypen	lookup	lookup table	2	6		<NA>	NaN	
Standorte	lookup	lookup table	2	5		<NA>	NaN	
OTBoxen	lookup	lookup table	1	7		<NA>	NaN	
OTSchnittKonzept	lookup	lookup table	1	4		<NA>	NaN	

table	group	purpose	rows	columns	key to mice	mice	% of all mice	tra
OTSchnittProzeduren	lookup	lookup table	1	6		<NA>	NaN	
Module	admin	menu entries of the web application	61	10		<NA>	NaN	
SystemVariablen	admin	configuration of the web application	14	3		<NA>	NaN	
Destinationen	admin	redirect targets of the web application	2	3		<NA>	NaN	

D2. Lab and sample tables

Coverage per trail (mice with at least one row), when the samples were taken relative to death, and the missingness per column.

trail	mice in trail	Gewebe	Blut	Knochenmark	Kryo	Paraffin	PCR
0	2	1	2	2	1	1	1
1	33	11	9	10	10	11	1
3	376	338	334	335	338	338	325
5	2	2	0	0	2	2	0
7	3	3	3	3	3	3	0
8	20	20	20	20	20	20	20
9	11	11	11	11	11	11	0
10	203	181	186	181	181	181	178
11	6	6	6	6	6	6	6
12	239	235	228	229	229	235	109
15	40	1	1	1	1	1	1
16	14	2	0	2	2	2	0
18	2	2	2	2	2	2	0

	rows with date and death date	% on the day of death (±1 day)	% earlier than 1 day before death	% later than 1 day after death	median days after death	date column
Gewebe	7362	99	1	0	-0.0	Entnahme
Blut	1028	100	0	0	-0.0	Entnahme
Knochenmark	1034	100	0	0	-0.0	Entnahme
PCR	3102	0	0	100	79.0	Datum

	kept series mice with a row	% of kept series mice
Gewebe	449	86.5
Blut	443	85.4
Knochenmark	448	86.3
Kryo	448	86.3
Paraffin	449	86.5
PCR	372	71.7

table	column	% NULL	% empty string	distinct values
Gewebe	**IdGewebe**	0.0	0.0	7362
	IdTier	0.0	0.0	813
	IdTumor	1.6	0.0	807
	IdTopographie	0.0	0.0	7
	IdSeite	1.6	0.0	4
	Gewebeart	0.1	0.0	18
	Entnahme	0.0	0.0	149
	Gewicht	1.7	0.0	217
	Bemerkung	0.0	8.0	221
	Auswertung	82.0	0.0	1
Blut	**IdBlut**	0.0	0.0	1028
	IdTier	0.0	0.0	802
	Entnahme	0.0	0.0	145
	LfdNr	0.0	0.0	207
	LfdNrTier	0.0	0.0	35
	Jahr	0.0	0.0	9
	Bemerkung	0.0	95.9	13
	Auswertung	40.6	0.0	1
	Type	78.2	0.0	1
Knochenmark	**IdKnochenmark**	0.0	0.0	1034
	IdTier	0.0	0.0	802
	Entnahme	0.0	0.0	145
	LfdNr	0.0	0.0	206
	LfdNrTier	0.0	0.0	35
	Jahr	0.0	0.0	9
	Bemerkung	0.0	99.7	4
	Auswertung	40.8	0.0	1
	Type	77.8	0.0	1
Kryo	**IdKryo**	0.0	0.0	3733
	IdTier	0.0	0.0	806
	IdTumor	3.2	0.0	799
	IdGewebe	0.0	0.0	3243

table	column	% NULL	% empty string	distinct values
	Probenart	0.0	0.0	9
	Tonne	0.0	10.7	4
	Turm	0.0	10.6	6
	Schublade	0.1	99.2	4
	Reihe	100.0	0.0	0
	Spalte	100.0	0.0	0
	IdFarbe	0.0	0.0	4
	LfdNr	0.0	0.0	209
	LfdNrTier	0.0	0.0	35
	Jahr	0.0	0.0	9
	Bemerkung	0.0	24.2	173
Paraffin	**IdParaffin**	0.0	0.0	4500
	IdTier	0.0	0.0	814
	IdTumor	0.0	0.0	808
	IdGewebe	0.0	0.0	4120
	Probenart	0.0	0.0	10
	Histokasten	51.0	0.8	228
	Tonne	100.0	0.0	0
	Schublade	100.0	0.0	0
	LfdNr	0.0	0.0	208
	LfdNrTier	0.0	0.0	36
	Jahr	0.0	0.0	9
	Bemerkung	0.1	19.1	197
PCR	**IdPCR**	0.0	0.0	3134
	IdTier	0.0	0.0	642
	IdGewKmBlut	0.0	7.4	1793
	GewebeArt	0.0	0.0	6
	Datum	0.0	0.0	67
	Zellzahl1	0.0	0.0	2518
	Zellzahl2	0.0	0.0	2541
	Zellzahl3	78.3	0.0	628
	MwZellen	0.0	0.0	2835

table	column	% NULL	% empty string	distinct values
	StAZellen	0.0	0.0	1645
	Cp1	0.0	0.0	1489
	Cp2	0.0	0.0	1492
	Cp3	78.3	0.0	422
	MwCp	0.0	0.0	2098
	StACp	0.0	0.0	356
	Bemerkung	0.0	78.0	10
	LfdNr	0.0	0.0	190
	LfdNrTier	0.0	0.0	36
	Jahr	0.0	0.0	9
	Sonderprobe	0.0	0.0	1

What the measured lab values look like: tissue weights per site (`Gewebe.Gewicht` , `Topographie`) and the ALU-PCR results (`PCR` , human cells per sample, by sample type).

site	rows	mice	with_weight	weight_g
Leber bei Sektionen	3249	802	3247	0.18 [0.10–0.30]
Lunge bei Sektionen	1601	803	1596	0.09 [0.06–0.11]
Großhirn bei Sektion	1598	800	1596	0.20 [0.18–0.21]
Primärtumor	796	734	796	0.33 [0.11–0.79]
Milz bei Sektion	115	115	0	–
Nieren bei der Sektion	2	1	2	0.00 [0.00–0.00]
Darmpaket bei der Sektion.	1	1	0	–

GewebeArt	rows	mice	trails	cells	Cp	first	last
blood	626	619	0, 1, 3, 8, 10, 11, 12, 15	0.1247 [0.0049–13.5288]	23.1 [21.6–26.8]	2014-01-16	2021-08-22
bm	618	613	0, 1, 3, 8, 10, 11, 12, 15	0.0208 [0.0007–1.4225]	25.6 [22.6–26.9]	2014-01-16	2021-08-24
brain	570	559	0, 3, 8, 10, 11, 12, 15	0.0016 [0.0000–0.3761]	25.5 [24.1–27.0]	2014-01-27	2021-08-25
liver	570	563	0, 3, 8, 10, 11, 12, 15	0.0128 [0.0004–0.6890]	25.7 [24.5–27.3]	2014-01-27	2021-08-25
lung	632	631	0, 3, 8, 10, 11, 12, 15	0.3962 [0.0026–19.6125]	22.1 [19.4–25.5]	2014-01-21	2021-08-23
spleen	118	112	11, 12, 15	0.0006 [0.0001–0.0025]	23.1 [22.7–24.0]	2021-09-13	2021-10-13

D3. `Logbuch`: an audit trail?

`Logbuch` is the log of the web application. Its `Bemerkung` column holds the raw request parameters, **including user names, session ids and passwords typed into the login form**, so it is never displayed here; only counts and the parameters of the visit forms are extracted.

Aktion	rows	first	last
show_login_site	72200	2020-05-08 06:21:56	2025-08-08 13:52:50
Querylog	52771	2020-05-08 06:28:40	2025-08-08 13:53:26
show_start_site_succ	48431	2020-05-08 06:29:47	2025-08-08 13:53:25
login_error	1101	2020-05-10 16:06:20	2025-08-08 13:52:53
login_succ	1063	2020-05-08 06:28:40	2025-08-08 13:52:56
logout_succ	716	2020-05-08 06:36:00	2025-06-23 05:30:36
show_start_site_error	637	2020-05-15 21:29:16	2024-12-02 17:53:17
1	2	2021-04-15 18:29:42	2021-04-15 18:29:45
fsh	1	2021-04-15 17:18:00	2021-04-15 17:18:00

```
Logbuch: 176,922 rows from 2020-05-08 to 2025-08-08; Querylog rows: 52,771; visit f
orm submissions with a weight: 23,013 new visits, 961 edits
```

Matching the logged visit forms with `Visiten`: a new-visit submission matches a stored visit of the same mouse at the same date and time (hh:mm); an edit names the stored visit (`visiteidcurr`).

	count
visits stored since the log starts	12168
... with a logged new-visit form (same mouse, date, hh:mm)	11564
... whose first logged weight differs from the stored weight	34
new-visit forms submitted more than once for the same mouse/date/time	8718
visits edited through the edit form (distinct visit ids)	331
... still in Visiten	331
... stored weight equals the last logged edit	330
... first entered weight known and different from the stored one	16

trail	visits	median_abs_change_g	max_abs_change_g
11	1	1.0	1.0
12	23	2.0	200.0
15	7	0.6	2.3
16	3	0.9	1.3

	trail	11	12	13	15	16	17	18
	visits	192	10062	2	1286	501	9	116
with_logged_form		140	9567	0	1243	489	9	116

D4. Old-table columns v1 never used

Share of visits per trail in which a column has content (not NULL, not empty and not 0, so a score of 0 counts as "no content"). The single score-sheet items (`AZ*`, `VH*`) are summarised as "any item > 0".

trail	visits	KGVerlauf (relative weight)	KGPunkte	any AZ item	AZPunkte	any VH item	VHPunkte	TUVolumen	TUVol
0	35	97.0	34.0	0.0	0.0	0.0	0.0	66.0	
1	1217	97.0	5.0	2.0	2.0	0.0	0.0	10.0	
2	7	0.0	0.0	0.0	0.0	0.0	0.0	100.0	
3	7047	95.0	5.0	1.0	1.0	0.0	0.0	43.0	
6	13	0.0	0.0	0.0	0.0	0.0	0.0	92.0	
7	39	92.0	0.0	8.0	8.0	0.0	0.0	77.0	
8	428	100.0	1.0	0.0	0.0	0.0	0.0	54.0	
9	11	0.0	0.0	0.0	0.0	0.0	0.0	100.0	
10	7159	97.0	7.0	0.0	0.0	0.0	0.0	70.0	
11	192	97.0	6.0	0.0	0.0	0.0	0.0	54.0	
12	10062	98.0	1.0	2.0	2.0	0.0	0.0	30.0	
13	2	50.0	0.0	0.0	0.0	0.0	0.0	100.0	
15	1286	98.0	3.0	0.0	0.0	0.0	0.0	25.0	
16	501	97.0	0.0	1.0	1.0	0.0	0.0	13.0	
17	9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
18	116	98.0	0.0	0.0	0.0	0.0	0.0	0.0	

trail	primary_tumour_records	with_weight	weight_g
0	1	1	0.66 [0.66–0.66]
1	10	3	0.28 [0.21–0.49]
2	6	0	–
3	393	270	0.16 [0.05–0.50]
5	2	2	0.49 [0.34–0.65]
7	3	0	–
8	40	40	0.36 [0.24–0.77]
9	11	11	1.27 [0.89–1.45]
10	211	172	0.89 [0.59–1.14]
11	6	6	1.16 [0.69–1.32]
12	241	217	0.24 [0.12–0.47]
13	2	0	–
15	40	1	0.20 [0.20–0.20]
16	7	2	0.38 [0.36–0.39]

Tagescheck (device check, not per mouse) rows per year: {2013: 57, 2014: 34, 2015: 7, 2016: 5, 2018: 18, 2019: 49, 2021: 37, 2022: 6, 2024: 9, 2025: 10}

What this tells us (Part D).

- Of the 30 new tables, 10 hold lab/sample data, 1 is the application log, 16 are lookups and 3 configure the web application.
- The lab tables are end-point data: Gewebe 99%, Blut 100%, Knochenmark 100% of the dated rows were taken within a day of death (necropsy); the `PCR.Datum` is the analysis date, a median of 79 days after death. Only `Gewebe.Gewicht` (organ/tumour weight) and the ALU-PCR cell counts are measurements; `Kryo`, `Paraffin`, `Blut` and `Knochenmark` mostly record where a sample is stored. They could be per-mouse **targets** (metastasis burden) or static covariates, not time-varying inputs for weight forecasting.
- `Logbuch` is a **web request log, not a row-level audit trail**: no table, no old value, no database user per change. It starts on 2020-05-08, so it can only cover trails visited since then. Its form submissions still show corrections: 11,564 of 12,168 visits stored since then (95%) have a logged entry form, 34 of them with a first weight different from the stored one (up to 200 g), and 331 visits were saved again through the edit form. It also stores plaintext passwords and should not be shared as is.
- Beyond weight, the per-visit columns with content are the total score (non-zero in trail 1 13%, trail 3 44%, trail 10 70%, trail 12 32%, trail 15 27%) and the tumour volume (> 0 in trail 1 10%, trail 3 43%, trail 10 70%, trail 12 30%, trail 15 25%); the single score-sheet items are almost never set (at most 2% of the visits in these trails) and `Belastung` is empty. Score and tumour volume are the candidate time-varying covariates.

Part E: data quality and summary

E1. Data-quality issues, all trails, both exports

v1's list (A5), extended by duplicate visits, visits after death and treatments without a visit on the same day.

unit	issue	old export	new export	added
mice	in trail 0, which is not in Versuchsreihen	2	2	0
	with role 0 (not in Rollen)	8	8	0
	without a death date	63	34	-29
	with an injection date before birth	2	2	0
visits	with an invalid date (e.g. 0000-00-00, day 00)	5	5	0
	of a mouse not in Versuchstiere	1	1	0
	before the birth of the mouse	29	29	0
	before the start of the trail	139	139	0
	more than 1 day after death	36	36	0
	with weight 0 g	10	10	0
	with weight > 100 g	11	11	0
	with weight between 0 and 10 g	1	1	0
	duplicates (same mouse, date and time)	142	188	46
	total score not the sum of the sub-scores	24	27	3
mouse-days	with more than one visit	304	352	48
treatments	RT fractions without a visit that day	27	27	0
	ChT administrations without a visit that day	23	23	0
	with an invalid date	0	0	0
trails	without a start date	1	1	0
	without mice	1	1	0

E2. Facts per export (as v1 Part D)

	old export (2023)	new export (2025)
Tables	25	55
Mice	843	988
Visits	22142	28125
Trails (Versuchsreihen)	15	18
Approved projects	12	15
Roles in Rollen	11	14
Visits: first – last date	2013-06-17 – 2022-04-07	2013-06-17 – 2025-06-24
RT fractions / ChT administrations / surgeries	1958 / 1316 / 47	2158 / 1406 / 67
Mice with a death date	780 (93%)	954 (97%)
Trail 3, RT/RChT/ChT through michael.load_dataset: mice / rows	186 / 3,049	186 / 3,049
All trails, RT/RChT/ChT: mice / rows	434 / 12,397	519 / 16,376
All trails, RT/RChT/ChT: trails with ≥ 10 mice	1, 3, 10, 12	1, 3, 10, 12, 15

E3. Candidate datasets for modelling

Concrete options, all from the new export through `michael.load_dataset` (v1's filters; the last option with the treatment filter switched off). Forecast origins are counted as in the rolling evaluation (`MIN_CONTEXT` = 2: every row from the third one of a mouse on).

	mice	rows	forecast origins	date span	trails	cell lines	% rows from trail 3	% rows with treatment	naive MAE, all origins (g)	c
1. Trail 3, RT/RChT/ChT (v1 study cohort)	186	3049	2677	2013-08 – 2016-08	1	3	100	26	0.662	experi 186 the 18 we 2016 give 4 the
2. Trails 3 + 10 + 12, RT/RChT/ChT	453	13757	12851	2013-08 – 2025-02	3	8	22	18	0.341	experi differe lines/s lines
3. All trails, RT/RChT/ChT	519	16376	15338	2013-06 – 2025-02	9	12	19	17	0.365	9 mar pro popul differ, tra
4. All trails, all roles, no treatment filter (context pool)	888	26974	25198	2013-06 – 2025-06	10	13	25	10	0.381	34% unt (co etc. as co not a

E4. Open questions for the data owners

1. **Kohorten.Versuchsreihe vs Versuchstiere.Versuchsreihe**: for 229 of 987 mice with a batch the trail of their cohort differs from their own trail (e.g. the TAS102 cohorts point to trail 8). Which one is right, or does `Kohorten.Versuchsreihe` mean something else?
2. **Trail 12 / PC-3DR**: the mice added to trail 12 come from cohort(s) PC-3DR RGB + Luziferase Hauptversuch (PC-3DR RGB + Luziferase). Is this a second experiment under the same trail id, and should it be analysed separately?
3. **Missing death dates**: 34 mice have no death date (by trail: 1: 2, 2: 7, 4: 6, 10: 8, 13: 2, 17: 9). Are these mice still alive, transferred, or are the records incomplete? Michael's loader drops them.
4. **Trail 0 and role 0**: 2 mice in trail 0 and 8 with role 0. What are they?
5. **Edited records**: 38 death records of trail(s) 15 were completed after 2023 and 4 chemotherapy routes changed from "unknown" to i.p. Are further corrections to the old data expected, and is there a change history before 2020-05-08 (when `Logbuch` starts)?
6. **Weight corrections**: 34 visits since 2020-05-08 were stored with a weight different from the one first entered in the form. Which value is right, and were older weights corrected without a trace?
7. **Logbuch** holds plaintext passwords and session ids. Should it be removed from exports given to us?
8. **Invalid dates**: 5 visits have dates like `0000-00-00` or day `00`. Can they be repaired?
9. **Lab samples**: what do `Blut.Auswertung` / `Knochenmark.Auswertung` ("ja") and `PCR.Sonderprobe` mean, and is the ALU-PCR cell count the intended metastasis read-out?

What this tells us (Part E).

- Most data-quality counts are the same in both exports (16 of 20 issues); the changes are without a death date mice: -29; duplicates (same mouse, date and time) visits: +46; total score not the sum of the sub-scores visits: +3; with more than one visit mouse-days: +48. The v1 problems "visits before the trail start" and "weights of 0 g / > 100 g" are unchanged, i.e. all in the old data.
- Candidate datasets range from 186 mice / 2,677 origins (trail 3) to 519 mice / 15,338 origins (all trails, RT/RChT/ChT) and 888 mice / 25,198 origins with untreated mice as context.
- Trails 3 + 10 + 12 hold 84% of the RT/RChT/ChT rows of all trails; the rest comes from trail 1 (1047 rows), trail 15 (900 rows), trail 16 (342 rows), trail 11 (181 rows), trail 8 (110 rows), trail 7 (39 rows).
- Naive last-value MAE differs by a factor of 4.0 between the kept trails (largest in trail 3, median gap 2 days; smallest in trail 10, median gap 1 day), so metrics on pooled data should also be reported per trail.